

Suheum Heo

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EDUCATION

University of Wisconsin-Madison, Madison, WI

May 2027

Bachelor of Science in Computer Science and Data Science

Relevant Coursework: Algorithms, Machine Learning, Database Management Systems, Theory & Algorithms for Data Science, Machine Organization & Programming, Data Science Modeling

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, Swift, SQL, R, HTML/CSS

Frameworks: React, Next.js, Node.js, FastAPI, Django/DRF, Flask, SwiftUI, Streamlit, Tailwind CSS

Data & ML: Pandas, NumPy, Scikit-learn, Plotly, Folium, DuckDB, Parquet, Jupyter Notebook

Databases & Tools: PostgreSQL, Supabase, Redis, Docker, Prisma, SQLAlchemy, Git/GitHub, Vercel

AI & APIs: Anthropic Claude API, REST APIs, WebSockets, Telegram Bot API, prompt engineering

TECHNICAL EXPERIENCE

Dassault Systèmes Korea - R&D Customer Success Intern | Seoul, South Korea

Jul-Aug 2026

- Developed an iOS proof-of-concept for an in-vehicle voice assistant integrating Korean speech-to-text, text-to-speech, and internal knowledge retrieval workflows.
- Built and physically tested a SwiftUI prototype with WebView and a mock service layer on iPhone; supported customer-facing R&D issue tracking, service requests, and cross-functional communication.

PROJECTS

Vibe Terminal - Crypto Trading Signal Terminal | FastAPI, PostgreSQL, Redis, React, WebSockets, Docker

Jul 2026

- Built a real-time crypto signal terminal monitoring Bithumb, Bybit, and KRW/USDT markets with an async FastAPI backend, WebSockets, Redis caching, PostgreSQL persistence, and Docker Compose.
- Engineered rule-based signal and risk workflows calculating VWAP, slippage, net premium, order-book depth, spread safety, and wallet status before Telegram alerts and position monitoring.

Madison Insights | Python, DuckDB, Streamlit, Parquet, Folium

May 2026

- Built a dual-source ETL pipeline ingesting 73,691 WI State Patrol crash records from historical ZIP/CSV files and WisTransPortal REST APIs with a concurrent 20-worker fetcher.
- Reconstructed missing highway names in 10,413 records and designed a severity-weighted SQL hotspot ranking; migrated to DuckDB + Parquet for zero-cost, sub-second Streamlit deployment.

FootyLens - Football Analytics Platform | FastAPI, Django/DRF, PostgreSQL, Redis, Next.js, Docker

May 2026

- Architected a health-checked 5-service platform for 20 Premier League clubs, separating async ingestion from Django ORM/auth and enforcing a 10 req/min token-bucket limit with Redis TTL caching.
- Built idempotent bulk synchronization eliminating 760+ DB round-trips per run and shipped xG, form, and top-scorer analytics through validated APIs and a server-rendered Next.js dashboard.

ScoutEdge - Football Transfer Intelligence | Next.js, TypeScript, Anthropic Claude API

Mar 2026

- Built an AI-assisted squad analysis and transfer-intelligence product combining football-data.org, API-Football, FotMob, Transfermarkt, and a local fuzzy-search database.
- Designed structured-output repair, prompt caching, fallback retries, and fact-grounding guardrails so tactical recommendations remain fast, resilient, and tied to verified player and club data.

MoneyMap - Personal Finance PWA | Next.js, TypeScript, Supabase, Tailwind CSS

May 2026

- Built and deployed an installable finance PWA tracking 500+ real expense entries with live currency conversion, Google OAuth, Supabase row-level security, real-time sync, and 7-language localization.

Common Ground - AI Classroom Discussion Tool | Node.js, PostgreSQL, Anthropic Claude API

Apr 2026

- Built a real-time Claude-powered discussion tool with QR-code joining and live results; selected as a finalist in the Anthropic x CDIS Speed-Hackathon.

ADDITIONAL EXPERIENCE

UW-Madison University Housing - Dining Operations | Madison, WI

2024-Present

- Cashier, Four Lakes Market (Dec 2025-Present): handle 200+ daily transactions. Previously Culinary Assistant & Shift Lead, Liz's Market (Nov 2024-Jul 2025): supervised 5-8 staff serving 500+ meals daily and trained 10+ employees.

Korea Army - Basic Training & Social Service Agent | Ulsan, South Korea

Oct 2022-Jul 2024

- Completed military training and supported elderly residents with dementia in a structured care environment, collaborating across diverse age groups with patience and clear communication.